

VertexLearn

Lecture: Introduction to Node.js

1. What is Node.js?

Node.js is a JavaScript runtime environment that allows JavaScript to run outside the browser. It is commonly used to build server-side applications, APIs, command-line tools, and real-time applications.

2. JavaScript Runtime

A runtime provides the environment required to execute a programming language. Node.js provides APIs and services that allow JavaScript programs to interact with the operating system, files, networks, and other resources.

3. The V8 JavaScript Engine

Node.js uses Google's V8 JavaScript engine. V8 converts JavaScript code into machine code and executes it efficiently. This engine is also used by the Chrome browser.

4. Event-Driven Architecture

Node.js follows an event-driven programming model. Applications can register callbacks for events and continue processing other work instead of waiting for every operation to finish.

5. Non-Blocking I/O

Node.js provides non-blocking input/output operations. For example, when a program reads a file or waits for a network response, it can continue handling other work while the I/O operation completes.

6. Event Loop

The event loop is a central part of Node.js asynchronous execution. It helps coordinate callbacks and completed asynchronous operations so that the application can continue processing tasks efficiently.

7. npm

npm, commonly known as the Node Package Manager, is used to install and manage packages for Node.js applications. Packages can provide reusable functionality such as web frameworks, database clients, authentication tools, and testing utilities.

8. Node.js Modules

Modules allow an application to organize functionality into separate files or reusable packages. Node.js applications can use built-in modules, third-party packages, and application-specific modules.

9. Common Use Cases

Node.js is commonly used for REST APIs, web servers, real-time applications, command-line tools, microservices, and backend services for web and mobile applications.

10. Simple Example

```
const http = require("http");

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js");
});

server.listen(3000);
```

11. Key Takeaways

Node.js runs JavaScript outside the browser, uses the V8 engine, supports event-driven and non-blocking I/O, provides an event loop for asynchronous processing, and has a large package ecosystem through npm.